

Mental Health Risk Prediction



Using CDC BRFSS 2024 Survey Data

Section A: Non-Technical Summary

Section B: Technical Overview

457,670

Respondents

37

Variables

5

Models Tested

OUR TEAM

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The Problem — Why Does Mental Health Matter?

1 in 5

U.S. adults experience mental illness
each year

15%

of our 457K respondents report frequent
mental distress

\$280B

estimated annual cost of mental illness in
the U.S.

Core Question

Can we use everyday survey data to predict who is most at risk — **BEFORE** they reach a crisis?



Early Detection

Identify high-risk individuals
before mental health
deteriorates to crisis level



Resource Allocation

Direct mental health services
toward groups who need
them most efficiently

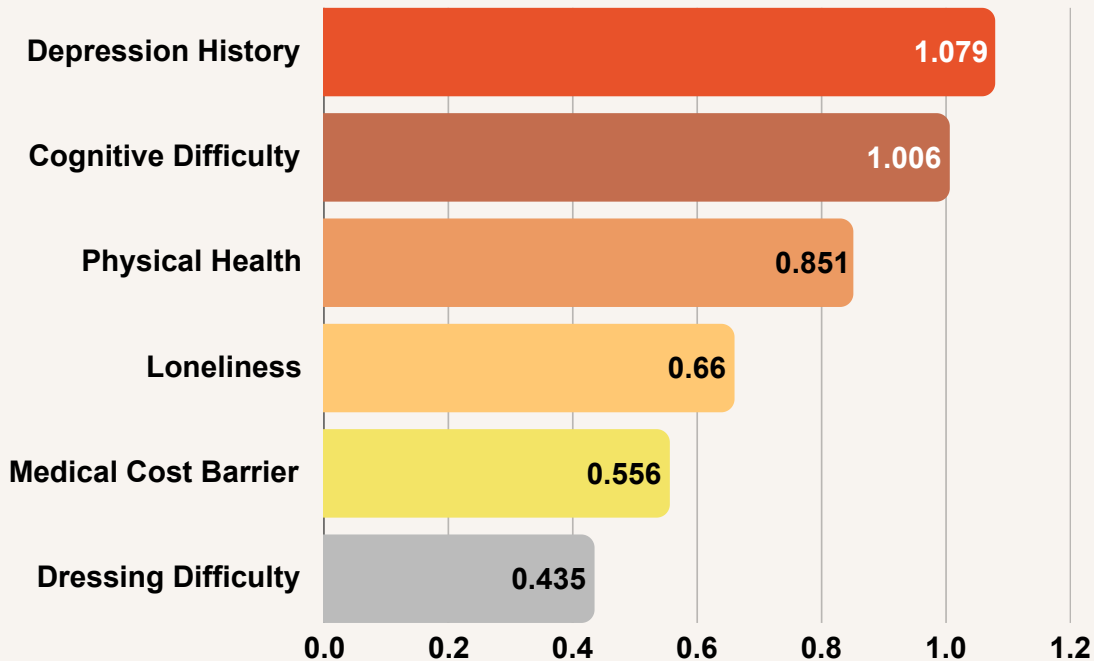


Policy Guidance

Give governments and
insurers evidence to design
better support programs

What Did We Find? — The 3 Biggest Predictors

QDA Standardized Class Mean Difference



#1

Depression History 1.08

Having ever been diagnosed with depression is by far the single strongest predictor.

#2

Cognitive Difficulty 1.01

People who struggle to concentrate or make decisions report far more bad mental health days.

#3

Physical Health 0.85

Body and mind are deeply linked — poor physical health days strongly predict poor mental health days.

What Do We Recommend?

For business leaders, healthcare providers, and policymakers

Healthcare Providers



Screen for Depression History First

Depression History alone accounts for 28% of predictive power. Any mental health screening program should make depression history the primary intake question.

Policymakers



Remove Financial Barriers to Care

People who can't afford doctor visits show significantly worse mental health outcomes. Subsidized access programs would directly reduce risk for the highest-need population.

Government & Insurers



Target Divorced & Low-Income Groups

EDA shows divorced, separated, and low-income individuals consistently rank as high-risk. Targeted outreach and community support programs could intervene early.

Healthcare Systems



Treat Physical & Mental Health Together

Physical health is the 3rd strongest predictor. Integrated care models that treat body and mind together will be more effective than siloed approaches.

Note: Our best model (Random Forest) explains ~33% of mental health variation. These are strong signals, not certainties — human judgment remains essential.

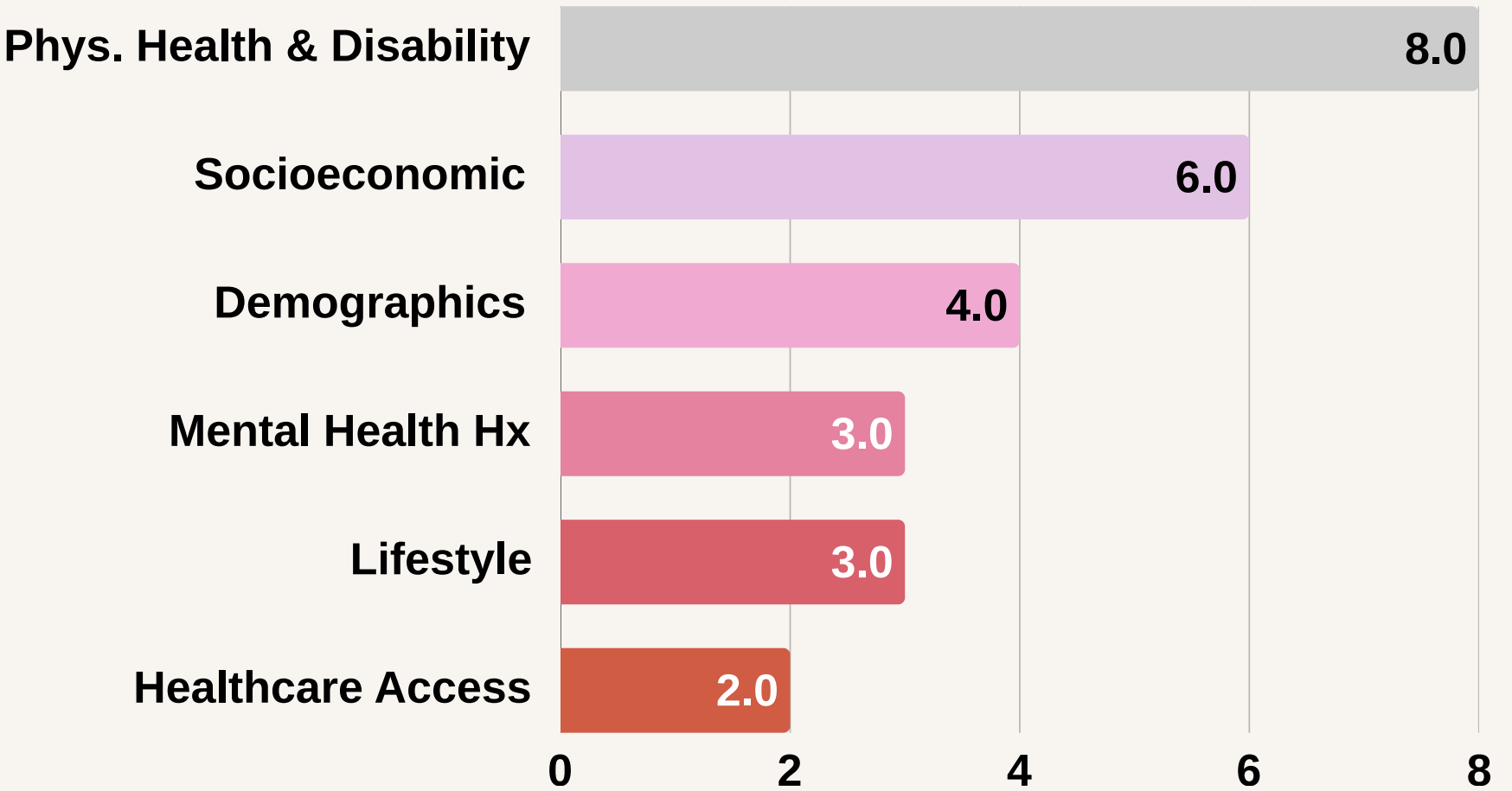
SECTION B

Technical Overview



Data — CDC BRFSS 2024

Behavioral Risk Factor Surveillance System · 457,670 respondents · 301 variables → 37 selected



Key Preprocessing Steps

01

Survey Code Recoding

77=Don't know, 99=Refused → NaN
88=None → 0 · Binary 2=No → 0

02

Column Removal

>200K missing (ACE series) → dropped
BMI engineered but dropped (r≈0.04)

03

Outlier Treatment

HEIGHT >7'11" → removed
WEIGHT ≥9000 (codes) → removed
0 duplicate rows found

Scenario A — Classification

Y = frequent_distress (1 if ≥14 bad mental health days, else 0) · 75/25 stratified split

Scenario B — Regression

Y = Mental Health (continuous 0–30 days of poor mental health) · 80/20 random split

Data — EDA

Interesting Finding in EDA



**Smoker
+ Can't Afford Dr.
= worst combo**



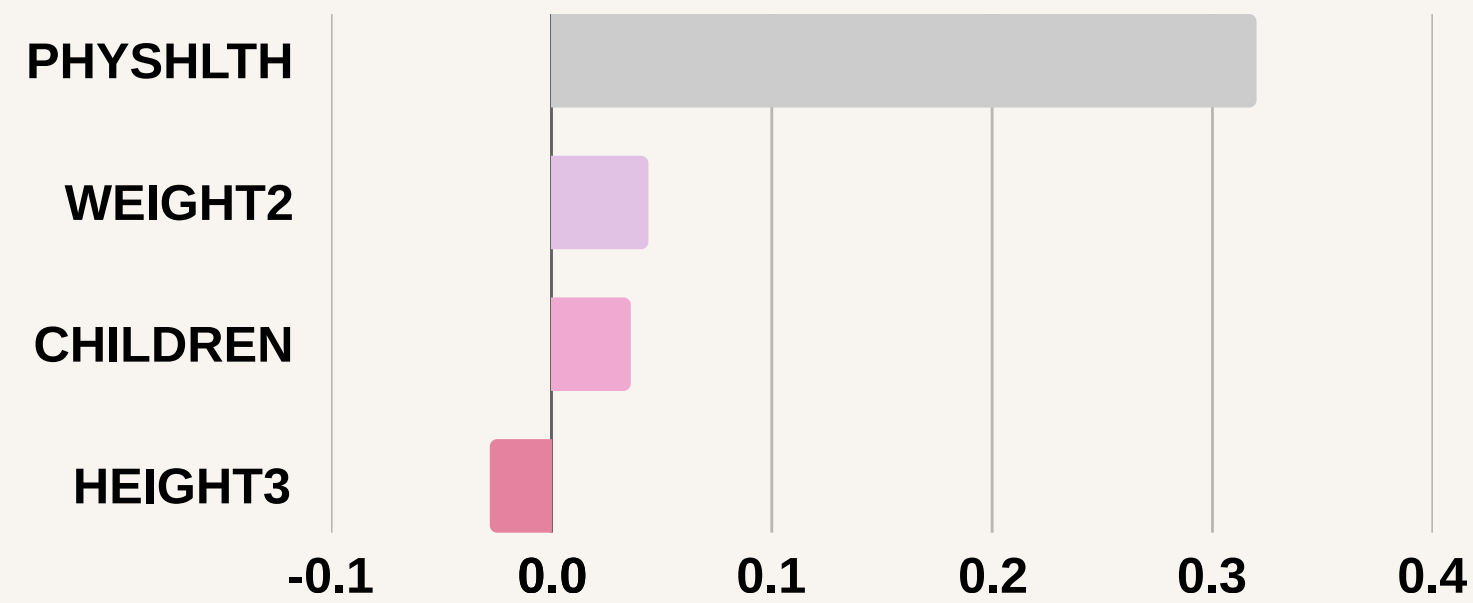
**Divorced women
> Divorced men**



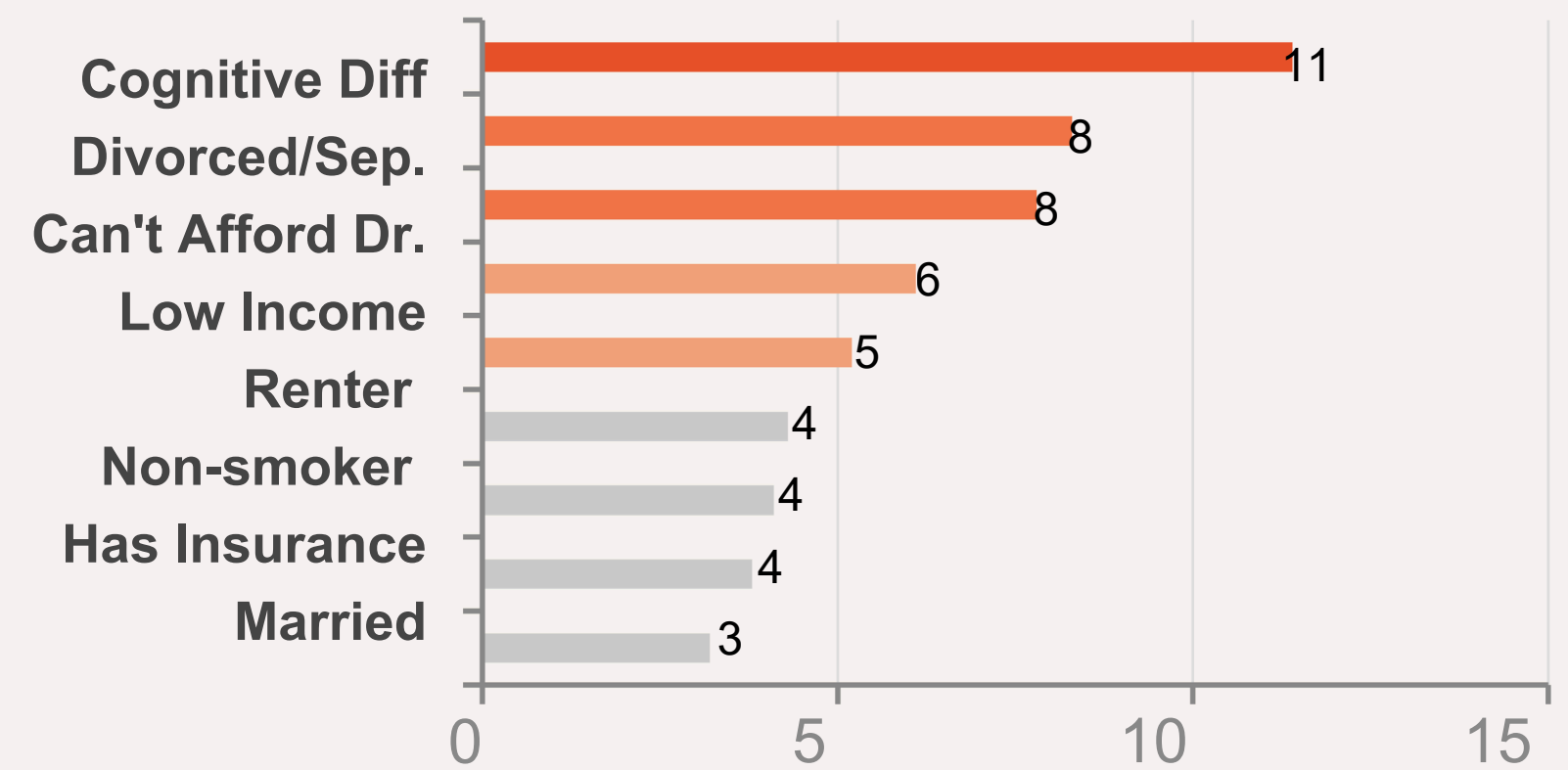
**Physical health vs
Mental health scatter**

EDA — Data Patterns & High-Risk Groups

Continuous Variable Correlations with MENTHLTH



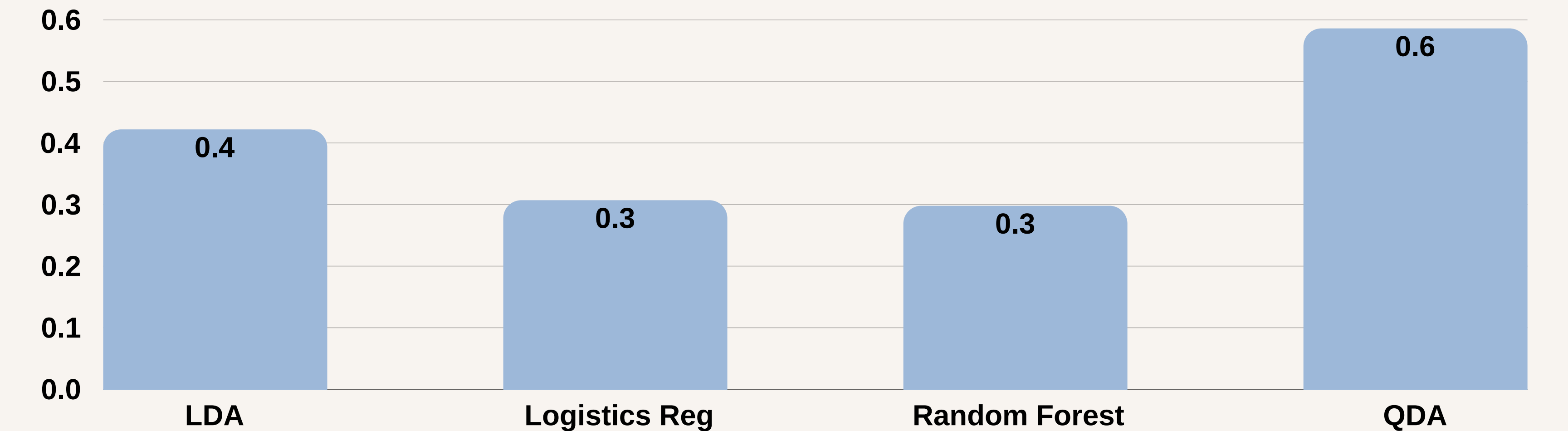
Avg Poor Mental Health Days by Group



3 Key EDA Takeaways

- Physical Health is the only continuous variable with meaningful correlation ($r=0.32$)
- Cognitive Difficulty group averages 11.4 days — 3.5× higher than the married group (3.2)
- 85/15 class imbalance → Accuracy is misleading; Precision-Recall metrics required

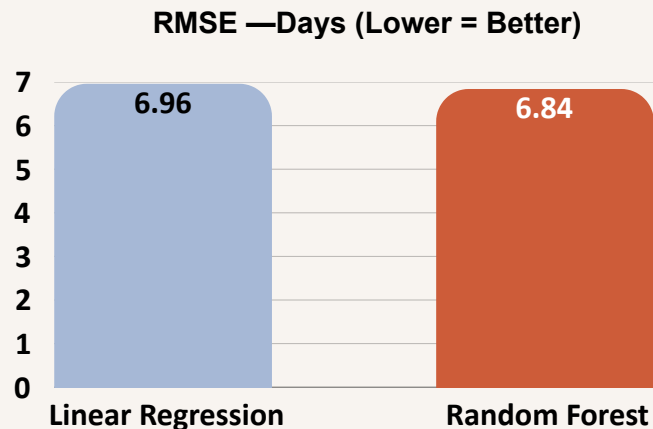
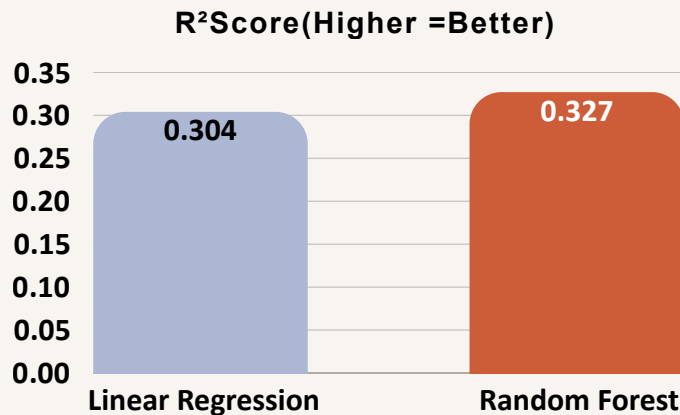
Results — Scenario A (Classification)



Model	Recall (Class 1)	ROC AUC	Precision (Class 1)	F1 (Class 1)	Best For
LDA	0.422	—	0.569	0.485	Baseline
QDA ★	0.586 ★	0.803	0.374	0.457	Best Recall — fewest missed cases
Logistic Reg.	0.307	0.837	0.650 ★	0.417	Best precision at threshold
Random Forest	0.298	0.846 ★	0.666	0.412	Best AUC — most reliable overall

Results —Scenario B (Regression)

Predicting the number of poor mental health days(0–30) · Metric: R^2 and RMSE



Interpreting the Results

- ✍ $R^2 = 0.327$ vs 0.100 : Random Forest explains 3×more variance than Linear Regression —non-linear interactions between variables matter.
- ✍ RMSE = 6.84 day.
- ✍ Random Forest captures complex variable interactions that a linear model simply cannot —justifying the added complexity.

Model Selection & Findings

Models Tested

LDA	Logistic Reg.	Linear Reg.	Random Forest	QDA
Scenario A · baseline	Scenario A · baseline	Scenario B · baseline	Scenario A + B	Scenario A · selected

What We Found

Top 3 predictors	Precision-Recall tradeoff	R ² ~0.30 (both models)
Depression · Cognitive Diff · PhysHealth Confirmed by both QDA class means and RF feature importance	QDA: best recall, lower precision RF: lower recall, best AUC	Mental health is genuinely complex ~33% variance explained by the model.

Thank You

3 Key Takeaways



01

Depression History Dominates

We prefer QDA for its 58.6% recall, driven mainly by depression history, cognitive difficulty, poor physical health, difficulty with daily tasks, and limited access to healthcare.

02

QDA Wins —But Mental Health Remains Complex

We prefer QDA for its high recall BUT:

*A reminder that prediction models complement, but cannot replace, human judgment.

03

Data tells the Real Story

Mental distress is not an isolated issue, but part of a broader pattern of overall health and socioeconomic vulnerability.